

NICHOLAS JOHNSON

Computational Neuroscience | Neuroinformatics

njohns59@gmu.edu | johnson.forum | linkedin.com/in/johnsonlink | github.com/nej296



EDUCATION

George Mason University

Expected Summer 2027

B.S. Neuroscience, Concentration: Computational Neuroscience | Fairfax, VA

- Curriculum focused in molecular/cellular neuroscience, computational neuroscience, neural networks, and neural informatics.
- Selected for GEO Neurotechnology in Germany Program, a 6-credit program visiting 8 research labs across Germany.
- Treasurer, Students in Neuroscience; administering Fall 2026 budget and expense tracking.

RESEARCH

NeuroMorpho.org Research Assistant

June 2026–Present

Neuroinformatics, Krasnow Institute for Advanced Study, George Mason University | Fairfax, VA

- Conduct literature mining to identify neuronal and glial reconstructions for NeuroMorpho.Org database curation.
- Evaluate reconstruction evidence using tracing-system keywords including Neurolucida, Imaris/Filament Tracer, Amira, Vaa3D, Simple Neurite Tracer, and CATMAID.
- Curate morphology metadata by verifying article details, reconstruction counts, figure evidence, supplementary files, data availability, and DOI/PMID references.
- Review repositories including GitHub, Dryad, Figshare, Zenodo, and Dataverse for downloadable reconstruction data.

Undergraduate Research Assistant

January 2026–May 2026

Anatomy and Physiology Education Research, George Mason University | Fairfax, VA

- Collaborated on manuscript preparation for a comparative anatomy and physiology education study.
- Contributed to data interpretation, scientific writing, editing, and narrative development for peer-reviewed publication.

PROJECTS

Hodgkin–Huxley Computational Neuroscience Project

May 2026–Present

Independent Research Project | Python-based biophysical neuron simulation with NEURON validation

- Designed and implemented a neuron simulator that uses the Hodgkin-Huxley equations to model action potential generation and membrane dynamics.
- Tested how injected current amplification parameters affect neuronal firing rates.
- Validated outputs against the NEURON Simulator; authored a research-style manuscript covering methods, error analysis, biological interpretation, and future extensions.

WORK EXPERIENCE

AI Trainer / LLM Evaluator

2026–Present

Handshake AI Fellowship | Remote

- Evaluated large language model responses for accuracy, reasoning quality, instruction-following, safety, and overall usefulness.
- Provided structured feedback on prompts and outputs to help improve AI assistant behavior across varied tasks and domains.

MRI Office Assistant

July 2025–Present

Institute for Biohealth Innovation, George Mason University | Fairfax, VA

- Support research administration office operations and completed entry-level MRI training from a certified MRI Technologist.

Office Assistant, IET Department

February 2025–May 2025

Northern Virginia Community College | Annandale Campus, VA

- Built computational computer-network labs, supported hardware setup, laptop debugging, and switch configuration for NOVA students using AI supported tools.

TECHNICAL SKILLS

Cellular & Molecular Neuroscience: Electrical Properties of Nerve cells, Synaptic Transmission, Neurotransmitters and their Receptors, Molecular Signaling within Neurons, Synaptic Plasticity

Computational neuroscience: NEURON Simulator; MOOSE; Github

Neuroinformatics: NeuroMorpho.org; ModelDB; literature mining; morphology metadata curation; reconstruction evidence review; DOI/PMID linking

AI evaluation: prompt development; LLM output review; factual accuracy assessment